SCARFACE: a harmonized spatio-temporal dataset integrating socio-economic, environmental, and agricultural indicators for the Po Valley (Italy), 2011–2024

Tables and list of available dataset
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Paolo Maranzano[1,2], Pietro Colombo[3], Felicetta Carillo[4], Riccardo Borgoni[1], Riccardo Pajno[1], Matteo Borrotti[1], Luca Ferrero[5] and Ezio Bolzacchini[5]

[1]University of Milano-Bicocca, Department of Economics, Management and Statistics (DEMS), Piazza dell'Ateneo Nuovo n.1, 20100, Milano, Italy

[2]Fondazione Eni Enrico Mattei, Corso Magenta n.63, 20123, Milano, Italy

[3]School of Mathematics and Statistics, University of Glasgow, UK

[4]Italian Council for Agricultural Research and Economics – Research Centre for Agricultural Policies and Bioeconomy (CREA-PB),

[5]University of Milano-Bicocca, Department of Earth and Environmental Sciences (DISAT)

Corresponding author: `paolo.maranzano@unimib.it`

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Abstract

We present *Sequestering CARbon through Forests, AgriCulture, and land usE (SCARFACE)*, a harmonized spatio-temporal dataset that integrates climate, air quality, airborne pollutant emissions, land cover, soil properties, agro-industry dynamics and socio-economic indicators, to jointly investigate interconnected processes linking agricultural systems, atmospheric dynamics, emissions, and socioeconomic conditions in the Po Valley, Northern Italy. The spatial reference unit is the Agrarian Sub Region, that is, groups of contiguous municipalities that are considered relatively homogeneous with respect to natural conditions, agronomic characteristics, and agricultural production systems. The dataset adopts an annual panel structure from 2011 to 2024 defined over the 256 ASRs partitioning the Po Valley and comprises more than 2,500 indicators sourced from national and international public institutions. Heterogeneous data are harmonized within a processing workflow, tailored to the specific characteristics of each dataset, that guarantee spatial and temporal consistency of the output dataset. The resource supports reuse in applied econometrics, spatio-temporal modeling, clustering, and policy analysis focused on agriculture, air quality, and land use in a major European hotspot.

Keywords: High-dimensional spatio-temporal dataset; Po Valley in Northern Italy; Agrarian regions; Agricultural activities; Air quality and pollution emissions; Meteorology; Spatial block kriging; Land cover; Livestock; Socio-economic indicators.

S1 Core thematic domains integrated in the SCARFACE dataset

Table 1: Overview of core thematic domains integrated in the SCARFACE dataset

Domain	**Data source**	**Spat. resolution**	**Temp. resolution**	**Period**	**ASR method**	**Num. indic.**
Farm activities	FADN (CREA)	Farm-level microdata	Annual	2011–2023	Population-weighted direct aggregation (sum/mean)	436
Emissions	EDGAR GHG v2025; EDGAR AP v8.1 (JRC, European Commission)	0.1° grid	Annual	2011–2024 (GHG); 2011–2022 (AP)	Spatial block kriging	96
Air quality (1)	European air quality interpolated data (EEA Datahub)	1 km grid	Annual	2011–2022	Spatial block kriging	8
Air quality (2)	CAMS European air quality reanalyses (CAMS Datahub)	0.1° grid	Hourly	2013–2024	Temporal aggregation + spatial block kriging	680
Meteorology	ERA5-Land reanalysis (ECMWF / Copernicus CDS)	0.1° hourly grid	Hourly	2011–2024	Temporal aggregation + spatial block kriging	1170
Extreme weather	European Drought Observatory (EDO)	0.1° grid	Infra-annual	2011–2024	Temporal aggregation + spatial block kriging	208
Land cover (1)	CORINE Land Cover (Copernicus)	100 m grid	Multi-year snapshots	2012 and 2018	Reclassification + piecewise constant + area shares	9
Land cover (2)	GDLC dataset (Global Dynamic Land Cover)	100 m grid	Multi-year snapshots	2015–2019	Reclassification + piecewise constant + area shares	6
Livestock	BDN livestock registry (Italian Ministry of Health)	Municipality (LAU)	Annual	2011–2024	Direct aggregation (sum)	16
Socio-economic indicators	Sistema informativo “A misura di comune” (ISTAT)	Municipality (LAU)	Annual	2014–2023	Direct aggregation (sum/mean)	95
Geographical features	Terrain elevation and morphology indicators (AWS Terrain Tiles; ISTAT)	Raster / administrative polygons	Static	Time-invariant	Spatial extraction + aggregation	10
Administrative metadata	NUTS territorial attributes (Eurostat GISCO)	Administrative regions	Static	Time-invariant	Spatial join to ASR	16

Note: All variables are harmonized at the ASR level through temporal aggregation, spatial block kriging, or direct aggregation depending on the nature of the original data source. The number of indicators (Num. Indic.) is given by the sum of annual and seasonal indicators (for CAMS, EDO and ERA5) or the sum of mean and total indicators plus the corresponding variances (for FADN data).

S2 Pollution emissions by sector and pollutant from EDGAR

Table 2: Agricultural emission sectors included in the SCARFACE dataset based on the EDGAR inventories. The table reports the corresponding IPCC classification codes and the GHGs and air pollutants considered for each sector.

Sector	Description	IPCC 1996	IPCC 2006	Emitted species
AGS	Agricultural soils	4C+4D1 4D2+4D4	3C2+3C3 3C4+3C7	N_2O, NH_3, NO_x NMVOC, PM_{10}, $PM_{2.5}$, CO_2, CH_4, N_2O,
AWB	Agricultural waste burning	4F	3C1b	CO, NO_x, NMVOC, PM_{10}, $PM_{2.5}$, BC, OC
ENF	Enteric fermentation	4A	3A1	CH_4, NMVOC, NH_3
MNM	Manure management	4B	3A2	CH_4, N_2O, NH_3, NMVOC, NO_x, PM_{10}, $PM_{2.5}$
N2O	Indirect N_2O emiss. from agriculture	4D3	3C5+3C6	N_2O

S3 Air quality indicators from EEA and CAMS

Table 3: Air quality datasets integrated in the SCARFACE database.

Dataset and source	**Spatial resolution** (grid)	**Temporal resolution**	**Pollutants (unit:** $\mu g\,/m^3$**)**
EEA interpolated air quality (European Environment Agency)	$1\,\mathrm{km} \times 1\,\mathrm{km}$	Annual	PM_{10}, $PM_{2.5}$, NO_2, NO_x
CAMS air quality reanalysis (Copernicus Atmosphere Monitoring Service)	$0.1° \times 0.1°$	Hourly → annual & seasonal	NH_3, $PM_{2.5}$, PM_{10}, NO_2, NO, SO_2, O_3, CO; secondary $PM_{2.5}$ components (ammonium, nitrate, sulfate, organic matter, elemental carbon); dust and sea salt

Table 4: Air pollutant variables included in the SCARFACE dataset

Pollutant	**Source**	**Temporal coverage**
PM_{10}	EEA	2011–2022
	CAMS	2013–2024
$PM_{2.5}$	EEA	2011–2022
	CAMS	2013–2024
NO_2	EEA	2011–2022
	CAMS	2013–2024
NO_x	EEA	2011–2022
NH_3	CAMS	2018–2024
CO	CAMS	2016–2024
SO_2	CAMS	2016–2024
O_3	CAMS	2013–2024
NO	CAMS	2018–2024
Dust	CAMS	2018–2024
Secondary inorganic aerosol	CAMS	2018–2024
$PM_{2.5}$ ammonium	CAMS	2023
$PM_{2.5}$ nitrate	CAMS	2023
$PM_{2.5}$ sulphate	CAMS	2023
$PM_{2.5}$ total organic matter	CAMS	2022–2024
$PM_{2.5}$ total elementary carbon	CAMS	2020–2024
$PM_{2.5}$ residential elementary carbon	CAMS	2020–2024
PM_{10} sea salt (dry)	CAMS	2022–2024
PM_{10} wildfires	CAMS	2019–2024

Note: Concentrations are measured in $\mu g/m^3$. Data are obtained from the CAMS European air quality reanalyses service: https://ads.atmosphere.copernicus.eu/datasets/cams-europe-air-quality-reanalyses?tab=overview (accessed on March 27th, 2026) or European Environmenta Agency (EEA) air quality data (interpolated data)- Series: https://www.eea.europa.eu/en/datahub/datahubitem-view/82700fbd-2953-467b-be0a-78a520c3a7ef (accessed on March 27th, 2026). CAMS data are originally provided ath hourly temporal resolutions; annual and seasonal (Winter, Spring, Summer, Fall) summary indicators (mean, minimum, maximum, and standard deviation) are computed prior to spatial aggregation. EEA data are already provided at the yearly frequency.

S4 Meteorological and land-surface variables from ERA5-Land and drought-related and extreme weather indicators from EDO

Table 5: Meteorological and land-surface variables derived from ERA5-Land hourly data and aggregated annually for Agrarian Sub Regions (ASRs) in Northern Italy (2011–2024). Available aggregation functions are minimum, mean, maximum and standard deviation. Seasons are defined according to climatic calendar, that is Winter (December–February), Spring (March–May), Summer (June–August), and Fall (September–November).

Variable	Description	Unit	Class	Notes
t2m	2m air temperature	°C	Atmospheric	Converted from Kelvin to Celsius before aggregation.
d2m	2m dew point temperature	°C	Atmospheric	Converted from Kelvin to Celsius before aggregation.
rh	Relative humidity	%	Atmospheric	Derived variable computed from air temperature ($t2m$) and dew point temperature ($d2m$) using the standard saturation vapor pressure formulation.
sp	Surface pressure	Pa	Atmospheric	
u10	10m zonal wind component	m/s	Atmospheric	
v10	10m meridional wind component	m/s	Atmospheric	
ws	Wind speed	m/s	Atmospheric	Derived from the horizontal wind components as $ws = \sqrt{u10^2 + v10^2}$.
wd	Wind direction	0-360°	Atmospheric	Derived from wind components using $wd = 180 - (\mathrm{atan2}(u10/ws, v10/ws) \cdot 180/\pi)$.
ssr	Surface solar radiation	J/m^2	Radiation	
str	Surface thermal radiation	J/m^2	Radiation	
tp	Total precipitation	mm	Hydrology	Converted from meters to millimeters before aggregation.
sf	Snowfall	mm	Hydrology	Converted from meters to millimeters before aggregation.
ro	Total runoff	mm	Hydrology	Converted from meters to millimeters before aggregation.
sro	Surface runoff	mm	Hydrology	Converted from meters to millimeters before aggregation.
ssro	Sub-surface runoff	mm	Hydrology	Converted from meters to millimeters before aggregation.
skt	Skin temperature	°C	Soil	Converted from Kelvin to Celsius before aggregation.
stl1	Soil temperature layer 1 (0–7 cm)	°C	Soil	Converted from Kelvin to Celsius before aggregation.
stl2	Soil temperature layer 2 (7–28 cm)	°C	Soil	Converted from Kelvin to Celsius before aggregation.

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Variable	Description	Unit	Class	Notes
stl3	Soil temperature layer 3 (28–100 cm)	°C	Soil	Converted from Kelvin to Celsius before aggregation.
stl4	Soil temperature layer 4 (100–289 cm)	°C	Soil	Converted from Kelvin to Celsius before aggregation.
swvl1	Soil volumetric water content layer 1	m^3/m^3	Soil	
swvl2	Soil volumetric water content layer 2	m^3/m^3	Soil	
swvl3	Soil volumetric water content layer 3	m^3/m^3	Soil	
swvl4	Soil volumetric water content layer 4	m^3/m^3	Soil	
cvh	High vegetation cover fraction	none	Vegetation	
cvl	Low vegetation cover fraction	none	Vegetation	
lai_hv	Leaf area index (high vegetation)	m^2/m^2	Vegetation	
lai_lv	Leaf area index (low vegetation)	m^2/m^2	Vegetation	

Table 6: Drought-related indicators and temperature extremes indicators included in the SCAR-FACE dataset

Variable	Description	Unit	Original temporal resolution	Temporal coverage
Soil moisture and drought indicators				
SMI	Normalized soil moisture index describing wet/dry conditions	/	Monthly	2011–2024
SMIAN	Anomaly of soil moisture index relative to climatology	/	Monthly	2011–2024
SMIAN-1	Short-term soil moisture anomaly (1-month scale)	/	Monthly	2011–2024
SMIAN-6	Medium-term soil moisture anomaly (6-month scale)	/	Monthly	2011–2024
SMIAN-12	Long-term soil moisture anomaly (12-month scale)	/	Monthly	2011–2024
CDI	Composite (combined) drought indicator integrating precipitation, soil moisture and vegetation response	Categorical index	Monthly	2012–2024
SPEI-1	Standardized Precipitation-Evapotranspiration Index (1-month scale)	Standardized index	Monthly	2011–2024
SPEI-6	Standardized Precipitation-Evapotranspiration Index (6-month scale)	Standardized index	Monthly	2011–2024
SPEI-12	Standardized Precipitation-Evapotranspiration Index (12-month scale)	Standardized index	Monthly	2011–2024
Temperature extremes indicators				
CWD	Cold wave duration: number of days with cold-wave conditions	Days	Daily	2011–2024
CWI	Cold Wave Intensity: intensity of cold-wave events	Index	Daily	2011–2024
HWD	Heat wave duration: number of days with heat-wave conditions	Days	Daily	2011–2024
HWI	Heat wave intensity: intensity of heat-wave events	Index	Daily	2011–2024

Note: Data are obtained from the European Drought Observatory (EDO), Copernicus Emergency Management Service: https://drought.emergency.copernicus.eu/tumbo/edo/download/ (accessed on March 27th, 2026). Original temporal resolutions vary across indicators (10-days, monthly, or daily) according to the specific product definition.

S5 Livestock consistency data from the National Livestock Registry (BDN)

Table 7: Livestock indicators derived from the National Livestock Registry (BDN) and included in the SCARFACE dataset.

Species	**Indicator**	**Livestock mode**
Bovine–buffalo	Number of farms	Stabled, SemiWild, Unknown
Bovine–buffalo	Number of heads	Stabled, SemiWild, Unknown
Swine	Number of farms	Stabled, SemiWild, Unknown
Swine	Number of heads	Stabled, SemiWild, Unknown

S6 Socio-economic indicators from ISTAT

Table 8: Socio-economic indicators included in the SCARFACE dataset.

Variable	Domain	Description	Unit	Aggregation
Pop_*	Demography	Resident population (by gender or age group)	persons	Sum
PopForeign_*	Demography	Foreign resident population (by gender)	persons	Sum
BirthRate	Demography	Birth rate	per 1,000 inhabitants	Mean
MortalityRate	Demography	Mortality rate	per 1,000 inhabitants	Mean
NetMigrRate	Demography	Net migration rate	per 1,000 inhabitants	Mean
OldAgeIndex	Demography	Old-age index	ratio	Mean
OldAgeDepIndex	Demography	Old-age dependency index	ratio	Mean
StructDepIndex	Demography	Structural dependency index	ratio	Mean
NrFam	Families	Number of families	households	Sum
NrFam_*	Families	Number of families by type	households	Sum
NrFamWithForeigners	Families	Families with foreign members	households	Sum
AvgFamilySize	Families	Average family size	persons per household	Mean
NrSingleIncomeFamWithYoungChildren	Families	Single-income families with children under 6	households	Sum
NrLowWorkIntensityFam	Families	Families with low work intensity	households	Sum
EmpRate	Labor market	Employment rate	%	Mean
UnempRate	Labor market	Unemployment rate	%	Mean
InactivityRate	Labor market	Inactivity rate	%	Mean
NEET1529	Labor market	Youth (15–29) not in employment, education or training	%	Mean
NrNonPermanentWorkers	Labor market	Non-permanent workers registered in October	persons	Sum
EntrepreneurshipRate	Economy	Entrepreneurship rate	%	Mean
LocalUnitsDensity	Economy	Density of local production units	units/km^2	Mean
LocationQuotients_*	Economy	Location quotient by economic sector	index	Mean
ShareLocalUnits_*	Economy	Share of local production units by sector	%	Mean
ShareWorkers_*	Economy	Share of workers by sector	%	Mean
HighTechSpecialization	Economy	High-tech production specialization index	index	Mean
TaxableIncomePerTaxpayer	Economy	Average taxable income per taxpayer	euros	Mean
NrVehicles	Transport	Registered passenger vehicles	vehicles	Sum
NrMotorcycles	Transport	Registered motorcycles	vehicles	Sum
RoadAccidentRate	Transport	Road accident rate	accidents per 1,000 inhabitants	Mean
RoadAccidentMortalityIndex	Transport	Road accident mortality index	index	Mean

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Variable	Domain	Description	Unit	Aggregation
RoadAccidentInjury Index	Transport	Road accident injury index	index	Mean
NrLibrariesPer100k	Culture	Libraries per 100,000 inhabitants	libraries per 100k inhabitants	Mean
CulturalSitesPer100k	Culture	Museums and cultural sites per 100,000 inhabitants	sites per 100k inhabitants	Mean
MuseumVisitorsPer100	Culture	Museum visitors	visitors per 100 inhabitants	Mean
SocialSpending PerCapita	Public finance	Social expenditure per capita	euros	Mean
RoadFinesRevenue PerCapita	Public finance	Road fines revenue per capita	euros	Mean
SpeedingFinesRevenue Share	Public finance	Share of fines from speed violations	%	Mean
WomenCouncilShare	Governance	Share of women in municipal councils	%	Mean
WomenExecutiveShare	Governance	Share of women in municipal executive boards	%	Mean
VoterTurnoutFirst Round	Governance	Voter turnout in municipal elections (first round)	%	Mean
AvgCouncilAge	Governance	Average age of municipal councillors	years	Mean
AvgAdministratorAge	Governance	Average age of municipal administrators	years	Mean
RecycleRate	Environment	Municipal waste recycling rate	%	Mean
LandConsumption	Environment	Land consumption	% of municipal area	Mean
SecondEduShare	Education	Share of population with secondary education	%	Mean
TertEduShare	Education	Share of population with tertiary education	%	Mean
NrChildcareChildren Served	Social services	Children enrolled in municipal childcare services	persons	Sum

Note: data are obtained from the experimental statistical dataset *Sistema informativo a misura di comune* (https://www.istat.it/statistica-sperimentale/aggiornamento-degli-indicatori-del-sistema-informativo-a-misura-di-comune/, accessed on April 3rd, 2026) produced by the Italian National Institute of Statistics (ISTAT). Available data cover the periodo 2014–2023. Variables ending with _* are further disaggregated by age group, gender, typology or sector (depending on the variable itself). ISTAT data are originally provided at the municipal level and then are aggregated at the ASR-level using the aggregation function reported in the last column.

S7 Land cover and land use variables from Copernicus

Table 9: Land-cover datasets integrated in the SCARFACE database.

Dataset	**Source**	**Classes / Resolution**	**Note**
CORINE Land Cover (CLC)	Copernicus Land Monitoring Service	44 classes; 100m	Data available for 2012 (CLC12) and 2018 (CLC18). Piecewise-constant assumption: CLC12 used for 2011–2017 and CLC18 for 2018–2024 under a piecewise-constant assumption
Global Dynamic Land Cover (GDLC)	Global land-cover dataset	GDLC classes; 100m	Data available for 2015-2019. Piecewise-constant assumption: the 2015 map was applied to 2011–2014 and the 2019 map to 2020–2024.

Table 10: Reclassification of CORINE Land Cover classes used in the SCARFACE dataset. Original CLC classes were grouped into nine broader land-cover categories according to the *_Reclassified nomenclature provided in the project mapping file.

Reclassified class	**Code Reclassified**	**Original CLC codes**	**Original CLC classes**
Urbanized area	`LC_CLC_Urban`	111, 112, 121, 122, 123, 124, 131, 132, 133, 141, 142	Continuous urban fabric; Discontinuous urban fabric; Industrial or commercial units; Road and rail networks and associated land; Port areas; Airports; Mineral extraction sites; Dump sites; Construction sites; Green urban areas; Sport and leisure facilities
Arable land	`LC_CLC_Arable-Land`	211, 212, 213	Non-irrigated arable land; Permanently irrigated land; Rice fields
Permanent crops	`LC_CLC_PermCrops`	221, 222	Vineyards; Fruit trees and berry plantations
Pastures	`LC_CLC_Pastures`	223, 231	Olive groves; Pastures
Heterogeneous agricultural areas	`LC_CLC_HetAgro`	241, 242, 243, 244	Annual crops associated with permanent crops; Complex cultivation patterns; Land principally occupied by agriculture, with significant areas of natural vegetation; Agro-forestry areas
Forests	`LC_CLC_Forests`	311, 312, 313	Broad-leaved forest; Coniferous forest; Mixed forest
Grassland, scrub and open spaces with little or no vegetation	`LC_CLC_GrassScrub OpenSpaceLilVeg`	321, 322, 323, 324, 331, 332, 333, 334, 335	Natural grassland; Moors and heathland; Sclerophyllous vegetation; Transitional woodland-scrub; Beaches, dunes, sands; Bare rocks; Sparsely vegetated areas; Burnt areas; Glaciers and perpetual snow
Wetlands	`LC_CLC_Wetlands`	411, 412, 421, 422, 423	Inland marshes; Peat bogs; Salt marshes; Salines; Intertidal flats
Water bodies	`LC_CLC_Water`	511, 512, 521, 522, 523	Water courses; Water bodies; Coastal lagoons; Estuaries; Sea and ocean

Table 11: Reclassification of Global Dynamic Land Cover (GDLC) classes used in the SCARFACE dataset. Original GDLC land-cover classes were harmonized into nine broader categories to ensure consistency with the CORINE Land Cover classification used in the project.

Reclassified class	**Code Reclassified**	**Original GDLC classes**
Urbanized area	`LC_GDLC_Urban`	Built-up areas and artificial surfaces
Arable land	`LC_GDLC_ArableLand`	Cropland; irrigated cropland; rainfed cropland
Permanent crops	`LC_GDLC_PermCrops`	Tree crops and plantations (e.g., orchards, vineyards)
Pastures	`LC_GDLC_Pastures`	Managed grasslands and pasture areas
Heterogeneous agricultural areas	`LC_GDLC_HetAgro`	Cropland–natural vegetation mosaics; mixed agricultural areas
Forests	`LC_GDLC_Forests`	Evergreen forest; deciduous forest; mixed forest
Grassland, scrub and open spaces with little or no vegetation	`LC_GDLC_GrassScrub OpenSpaceLilVeg`	Natural grasslands; shrublands; sparse vegetation; barren land
Wetlands	`LC_GDLC_Wetlands`	Inland wetlands; marshes; peatlands
Water bodies	`LC_GDLC_Water`	Rivers; lakes; reservoirs; coastal water bodies

S8 Time-invariant geographical features

Table 12: Time-invariant geographical features included in the SCARFACE dataset.

Feature	**Description**	**Source**
Area (m^2)	Total land area of the Agrarian Sub Region computed from the ASR polygon geometry (in square meters).	ASR geometries
Elevation (m)	Minimum, maximum, mean, and standard deviation of elevation computed within each ASR from a digital elevation model with approximately 30 m ground resolution.	AWS Terrain Tiles
Share % Area (Plain)	Share of ASR area located at elevation $\leq$ 200 m.	DEM derived indicators
Share % Area (Hills)	Share of ASR area located at elevation between 200 m and 600 m.	DEM derived indicators
Share % Area (Mount)	Share of ASR area located at elevation $>$ 600 m.	DEM derived indicators
Longitude, Latitude	Geographic coordinates of the centroid of each ASR polygon.	ASR geometries
NUTS administrative attributes	Hierarchical identifiers and territorial classifications for NUTS0–NUTS3 regions, including urban–rural typology, remoteness, metropolitan status, coastal and mountain indicators, border status, and land surface measures.	Eurostat GISCO

S9 Techno-economic specialization of farms and agronomic practices from FADN

Table 13: Farm Accountancy Data Network (FADN) variables included in the SCAR-FACE dataset

Variable	Domain	Description	Unit
Bio	Agronomic practices	Share of organic farms in the ASR	0–100%
Fert_NitrogenTot	Agronomic practices	Fertilizers – Total quantity of nitrogen employed	Quintals
Fert_PhosphorusTot	Agronomic practices	Fertilizers – Total quantity of phosphorus employed	Quintals
Fert_PotassiumTot	Agronomic practices	Fertilizers – Total quantity of potassium employed	Quintals
FertFertHectares	Agronomic practices	Fertilizers – Surface on which fertilizers are spread	Hectares
Fert_NitrogenPerHA	Agronomic practices	Fertilizers – Nitrogen per hectare	Quintals per ha
Fert_PhosphorusPerHA	Agronomic practices	Fertilizers – Phosphorus per hectare	Quintals per ha
Fert_PotassiumPerHA	Agronomic practices	Fertilizers – Potassium per hectare	Quintals per ha
Phyto_QtyPerHA_ToxicClass0	Agronomic practices	Phytopharmaceuticals (class 0 – non-toxic) per hectare	kg per ha
Phyto_QtyPerHA_ToxicClass1	Agronomic practices	Phytopharmaceuticals (class 1 – slightly toxic) per hectare	kg per ha
Phyto_QtyPerHA_ToxicClass2	Agronomic practices	Phytopharmaceuticals (class 2 – moderately toxic) per hectare	kg per ha
Phyto_QtyPerHA_ToxicClass3	Agronomic practices	Phytopharmaceuticals (class 3 – highly toxic) per hectare	kg per ha
Phyto_QtyPerHA_ToxicClass4	Agronomic practices	Phytopharmaceuticals (class 4 – extremely toxic) per hectare	kg per ha
Phyto_QtyTot_ToxicClass0	Agronomic practices	Total quantity of phytopharmaceuticals (class 0 – non-toxic) employed	kg
Phyto_QtyTot_ToxicClass1	Agronomic practices	Total quantity of phytopharmaceuticals (class 1 – slightly toxic) employed	kg
Phyto_QtyTot_ToxicClass2	Agronomic practices	Total quantity of phytopharmaceuticals (class 2 – moderately toxic) employed	kg
Phyto_QtyTot_ToxicClass3	Agronomic practices	Total quantity of phytopharmaceuticals (class 3 – highly toxic) employed	kg
Phyto_QtyTot_ToxicClass4	Agronomic practices	Total quantity of phytopharmaceuticals (class 4 – extremely toxic) employed	kg
Phyto_Hectares_ToxicClass0	Agronomic practices	UAA surface where class 0 (non-toxic) phytopharmaceuticals are applied	ha
Phyto_Hectares_ToxicClass1	Agronomic practices	UAA surface where class 1 (slightly toxic) phytopharmaceuticals are applied	ha
Phyto_Hectares_ToxicClass2	Agronomic practices	UAA surface where class 2 (moderately toxic) phytopharmaceuticals are applied	ha
Phyto_Hectares_ToxicClass3	Agronomic practices	UAA surface where class 3 (highly toxic) phytopharmaceuticals are applied	ha
Phyto_Hectares_ToxicClass4	Agronomic practices	UAA surface where class 4 (extremely toxic) phytopharmaceuticals are applied	ha

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Variable	Domain	Description	Unit
ProdSO	Farms structure	Agricultural Standard Output: value of the potential production generated from LSU and UAA	Ths. €
TotalLaborHours	Farms structure	Total hours worked in a year	Hours
LSU	Farms structure	Livestock Standard Unit	Units
UAA	Farms structure	Utilized Agricultural Area (hectares)	ha
AWU	Farms structure	Annual Work Unit: total hours worked divided by 1,800 (employees) and 2,200 (family labor)	Units
ManureQL_QtyInvInit	Agronomic practices	Total quantity of manure (stock) at January 1st	Quintals
ManureQL_ValInvInit	Agronomic practices	Value of manure (stock) at January 1st	€
ManureQL_QtyInvFin	Agronomic practices	Total quantity of manure (stock) at December 31st	Quintals
ManureQL_ValInvFin	Agronomic practices	Value of manure (stock) at December 31st	€
ManureQL_QtySales	Agronomic practices	Quantity of manure sold during the year	Quintals
ManureQL_ValSales	Agronomic practices	Value of manure (stock) sold during the year	€
ManureQL_QtyProd	Agronomic practices	Quantity of manure produced during the year	Quintals
ManureQL_QtyOther	Agronomic practices	Quantity of manure employed in other ways during the year	Quintals
ManureQL_ValOther	Agronomic practices	Value of manure (stock) employed in other ways during the year	€
Machine_Number	Agronomic practices	Number of machines	Units
Machine_TotPowerKW	Agronomic practices	Total power of available agricultural machinery	kW
Machine_AvgPowerKW	Agronomic practices	Average power per machine	kW per machine
Machine_Hours	Agronomic practices	Usage hours of agricultural machinery	Hours
CarbFoot	Environmental impact	Agricultural carbon footprint	Ths. tonnes
RevenuesTotal	Econ. bal.: revenues	Total revenue	€
FGSP_Total	Econ. bal.: revenues	Total Farm Gross Saleable Production (FGSP)	€
FGSP_TransfProducts	Econ. bal.: revenues	FGSP from processed products	€
FGSP_DirectSales	Econ. bal.: revenues	FGSP from direct sales of agricultural products	€
FGSP_Crops	Econ. bal.: revenues	FGSP from crops	€
FGSP_Livestock	Econ. bal.: revenues	FGSP from livestock	€
FGSP_RenewEnergy	Econ. bal.: revenues	FGSP from renewable energy produced by the farm	€
FGSP_Quality	Econ. bal.: revenues	FGSP from high-quality products	€
SalesProducts	Econ. bal.: revenues	Revenue from sales of products and services	€
InventoryChange	Econ. bal.: revenues	Change in inventories	€
AidsEU	Econ. bal.: subsidies	Operating public grants from EU CAP policies (Pillar I)	€
Selfconsumes	Econ. bal.: revenues	Self-consumption, gifts, and in-kind wages	€
IncreasePhysAssets	Econ. bal.: revenues	Capitalized internal work (internally generated assets)	€
RevenuesAgritourism	Econ. bal.: revenues	Agritourism revenue	€

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Variable	Domain	Description	Unit
RevenuesThirdPartyServices	Econ. bal.: revenues	Contract work revenue (custom farming services)	€
RevenuesComplementary	Econ. bal.: revenues	Complementary activities revenue	€
RentActive	Econ. bal.: revenues	Rental income (active leases)	€
CurrentCosts	Econ. bal.: costs	Total current costs	€
Seeds	Econ. bal.: costs	Seeds and seedlings costs	€
Fertilizers	Econ. bal.: costs	Fertilizer costs	€
PesticidesHerbicides	Econ. bal.: costs	Pesticides and herbicides costs	€
AnimalFeed	Econ. bal.: costs	Animal feed costs	€
ForageBedding	Econ. bal.: costs	Forage and bedding costs	€
Machines	Econ. bal.: costs	Mechanization costs	€
Utilities	Econ. bal.: costs	Water, electricity, and fuel costs	€
ConsFactorsAgritourism	Econ. bal.: costs	Agritourism input costs	€
ExpenditureCommercialTransform	Econ. bal.: costs	Processing, marketing, and storage costs	€
ExpenditureGeneraliFondiarie	Econ. bal.: costs	General and land-related overheads	€
ThirdPartyServices	Econ. bal.: costs	Third-party services costs	€
RentPassive	Econ. bal.: costs	Equipment rental costs (operating leases)	€
VeterinaryExpenditure	Econ. bal.: costs	Veterinary and health costs	€
AgritourismCosts	Econ. bal.: costs	Agritourism service costs	€
Insurance	Econ. bal.: costs	Insurance costs	€
ValueAdded	Econ. bal.: results	Value added: total revenue – current costs	€
MultiYearCosts	Econ. bal.: costs	Multi-year costs (depreciation and provisions)	€
Depreciation	Econ. bal.: costs	Depreciation	€
Provisions	Econ. bal.: costs	Provisions	€
FarmNetIncome	Econ. bal.: results	Net farm income (value added – multi-year costs)	€
DistributedIncome	Finance	Distributed income (wages, social security, rent)	€
SalariesSocialCosts	Econ. bal.: costs	Wages and social security contributions	€
RentsPassive	Econ. bal.: costs	Rent expenses (passive leases)	€
OperativeIncome	Econ. bal.: results	Operating income	€
NonOperatingIncome	Econ. bal.: results	Non-operating income	€
FinancialRevenues	Econ. bal.: revenues	Financial income	€
FinancialCosts	Econ. bal.: costs	Financial expenses	€
AidsPublicCapitalAccount	Econ. bal.: subsidies	Capital grants (public investment subsidies)	€
OtherAids	Econ. bal.: subsidies	Non-EU operating subsidies	€
CurrentTaxes	Econ. bal.: costs	Current taxes	€
NetIncome	Econ. bal.: results	Net income	€
VariableCosts	Econ. bal.: costs	Variable costs	€
FixedCosts	Econ. bal.: costs	Fixed costs	€
GrossOperatingMargin	Econ. bal.: results	Gross operating margin (value added – labor costs)	€
FarmNetValueAdded	Econ. bal.: results	Farm net value added	€
Aids_EUPillarIprod_Amount	Econ. bal.: subsidies	EU CAP subsidies (Pillar I – production)	€
Aids_EUPillarIIrurdev_Amount	Econ. bal.: subsidies	EU CAP subsidies (Pillar II – rural development)	€
Aids_StateRegLocal_Amount	Econ. bal.: subsidies	National, regional, and local subsidies	€
AvgFGSPcrops_UAA	Productivity	Average FGSP from crops per hectare of UAA	€per ha

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Variable	Domain	Description	Unit
AvgFGSPlivestock_LSU	Productivity	Average FGSP from livestock per LSU	€per LSU

Note: data are obtained from the Italian Farm Accountancy Data Network (FADN) survey (https://www.crea.gov.it/en/web/politiche-e-bioeconomia/-/fadn-farm-accountancy-data-network), co-ordinated by the Italian Council for Agricultural Research and Agricultural Economics Analysis (CREA). For every variables, yearly farm-level values are aggregated at the ASR-level using the Horvitz-Thompson estimator of the mean and the total, both augmented by the corresponding estimate of the area-level variance. All the variables cover the period 2011–2023 with time-varying spatial coverages (i.e., the list of areas covered can change across the years). Variables are classified according to thematic domains: agronomic practices, farms structure, economic balanca (revenues, costs, results, and subsidies) and productivity.